Chapter 2 Breakout (Pt. III)

STAT 4714

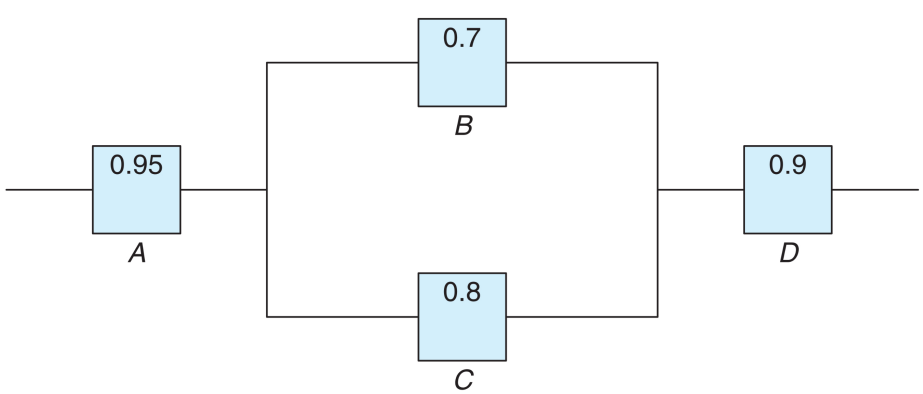
Questions

1. An ECE class is composed of 10 juniors, 30 seniors, and 10 graduate students. The final grades show that 3 of the 10 juniors, 10 of the seniors, and 5 of the graduate students received an A for the course. If a student is chosen at random and is found to have earned an A, what is the probability that they are a senior?

2. The probability that a doctor correctly diagnoses a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient files a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues?

3. A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.

   (a) What is the probability that neither is available when needed?

   (b) What is the probability that a fire engine is available when needed?

4. Suppose the diagram of an electrical system is given as below. Assume that the components fail independently. What is the probability that the system works?